

Ejercicio 1: Aplicando la Transformada Z, y utilizando la **propiedad de desplazamiento en el tiempo**, determine la función de transferencia $H(z)$ de los siguientes sistemas LTI causales:

Si $x[n] \leftrightarrow X(z)$
 $\Rightarrow x[n-n_0] \leftrightarrow X(z) \cdot z^{-n_0}$

$$H(z) = \frac{Y(z)}{X(z)}$$

1. $y[n] - \frac{1}{2}y[n-1] + \frac{1}{4}y[n-2] = x[n]$
2. $y[n] = y[n-1] + y[n-2] + x[n-1]$
3. $y[n] = 7x[n] + 2y[n-1] - 6y[n-2]$
4. $y[n] = \sum_{k=0}^7 2^{-k}x[n-k]$

1.

$$y[n] - \frac{1}{2}y[n-1] + \frac{1}{4}y[n-2] = x[n]$$

$$Y(z) - \frac{1}{2}Y(z)z^{-1} + \frac{1}{4}Y(z)z^{-2} = X(z)$$

$$H(z) \Rightarrow Y(z) \left[1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2} \right] = X(z) \cdot 1$$

$$\frac{Y(z)}{X(z)} = \frac{1}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}} = H(z)$$

Función de transferencia del sistema.
 Esta $H(z)$ es la transformada z a la respuesta del impulso $h[n]$ del sistema.

2.

$$2. y[n] = y[n-1] + y[n-2] + x[n-1]$$

$$Y(z) - Y(z)z^{-1} - Y(z)z^{-2} = X(z)z^{-1}$$

$$Y(z) \left[1 - z^{-1} - z^{-2} \right] = X(z) \cdot \left[1 \cdot z^{-1} \right]$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z^{-1}}{1 - z^{-1} - z^{-2}}$$

3.

$$3. y[n] = 7x[n] + 2y[n-1] - 6y[n-2]$$

$$Y(z) - 2Y(z)z^{-1} + 6Y(z)z^{-2} = 7X(z)$$

$$Y(z) \left[1 - 2z^{-1} + 6z^{-2} \right] = X(z) \left[7 \right]$$

$$\frac{7}{1 - 2z^{-1} + 6z^{-2}} = H(z) \Leftarrow$$

4.

$$4. y[n] = \sum_{k=0}^7 2^{-k}x[n-k]$$

$$Y(z) = \sum_{k=0}^7 2^{-k}X(z)z^{-k}$$

$$H(z) = \sum_{k=0}^7 2^{-k}X(z)z^{-k}$$

$$H(z) = \sum_{k=0}^{\infty} 2^{-k} x(z) z^{-k}$$